COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
DATE PERFORMED:	September 16, 2026	**DATE SUBMITTED:**	September 16, 2026
NAME:	Silao, Johrel Louie T.	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

ACTIVITY NO. _9.1_ Implementing Trees

1. PROCEDURE OUTPUT

- Submit a document explaining the methods/functions of the **BinaryTree** class you created.

1. **Class BinaryTree**
   - It is a node based design where each instance represents an individual node and the root of its own subtree.

2. **Constructor BinaryTree**
   - This initializes a tree node with a value and sets both left and right child to the nullptr.

3. **insertLeftChild**
   - Allocates a new node that serves as a left child and if this exists, it pushes the existing child down one level to become the left child of the newly inserted node.

4. **insertRightChild**
   - Allocates a new node that serves as a right child and if this exists, it pushes the existing child down one level to become the right child of the newly inserted node.

5. **getLeftChild**
   - Returns a pointer to the left subtree node.

6. **getRightChild**
   - Returns a pointer to the right subtree node.

7. **getRootVal**
   - Returns the stored value stored in the current node.

8. **setRootVal**
   - Updates the stored value of the current node to obj.

9. **isLeaf**
   - Verifies to true if left and right child are nullptr that indicates if the node has no children.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.